

DECO2300/7230 – Final Report and Reflection

Project: Procreate-style XR Creative Tool (Meta Quest)

Author: Yucheng Chen

1. Objective and Validation Metrics

The goal of this project was to create an immersive XR environment where users can draw, erase, and manage creative tools in a natural and responsive way.

Each prototype stage built on the previous one to test interaction flow, physical realism, and user comfort.

Prototype 1 explored how users understood the workflow on a simple desktop interface.

Prototype 2 introduced the main XR functions, such as 3D drawing, remote grabbing, and a physical reset button.

Prototype 3 added physical realism with drawers, levers, and height adjustment to study immersion.

The key objectives for testing were:

- To evaluate how realistic and smooth the drawing and erasing actions felt in VR.
- To test the reliability and clarity of the physical reset button and wrist menu.
- To examine whether feedback (sound, haptic, and visual) helped users stay confident during tasks.
- To measure overall ease of learning and comfort.

Success criteria

- At least 80% task completion rate.
- Average satisfaction ≥ 4 out of 5.
- Gestures and controls accurate and intuitive.
- No major mechanical or visual errors.

These criteria guided the evaluation and helped compare progress across all prototypes.

2. Results

● Prototype 1 – Desktop Simulation

Prototype 1 tested the basic creative workflow using mouse and keyboard.

All five users completed all tasks with an **average satisfaction of 4.4/5**.

Drawing and erasing felt simple and natural.

Users liked the reset button for quick correction but asked for clearer visual hints, sound

feedback, and undo/redo options.

This stage confirmed that the creative flow was clear, but stronger visual and audio feedback was needed for connection and control.

● **Prototype 2 – Core XR Prototype (Main Focus)**

Prototype 2 introduced full VR drawing, dual erasers, a reset button, a wrist menu, and remote grabbing.

All users finished all tasks, with **average satisfaction around 4.3/5**.

Drawing and erasing felt realistic, especially the soft eraser.

The **remote grab and floating tools** received perfect ratings (5/5) for improving speed and immersion.

The **reset button** worked but felt stiff and sometimes lost sound feedback.

The **wrist menu** opened easily but lacked tool functions.

Minor canvas penetration reduced realism but did not stop task success.

Prototype 2 validated all major XR interactions.

It proved that the creative experience was engaging and intuitive, though feedback consistency and button mechanics needed refinement.

● **Prototype 3 – More Physical Interactions**

Prototype 3 added a drawer, shutter door, lever, and camera height control to test realism.

All users rated realism and responsiveness **4–5/5**.

They enjoyed the lever and drawer, but some wanted smoother motion, light resistance, and matching sound or vibration.

A few asked for visual highlights to locate interactable parts.

This stage showed that physical realism improved immersion and comfort but also required balanced motion and better feedback cues.

3. Analysis / Insights

Prototype 1 showed that users could easily understand the drawing workflow.

All participants completed tasks smoothly, confirming that the basic creative logic was clear.

However, users mentioned that actions felt “flat” without sound or visual cues.

This revealed that early feedback design was too minimal and that sensory responses are essential for maintaining engagement and confidence.

The main lesson from this stage was that simple interactions can still feel incomplete without feedback that connects users to the system.

Prototype 2 provided the strongest insights into core XR performance.

Drawing and erasing were accurate and responsive, while the **remote grab and floating tools**

stood out as the most successful feature. They reduced unnecessary arm movement and supported continuous creative flow, directly improving user efficiency and satisfaction. In contrast, the **reset button** and **wrist menu** exposed weaknesses in reliability and feedback. The button’s inconsistent sound and stiffness broke immersion, and the wrist menu, though well-placed, lacked active tools. This phase proved that functional feedback and clear affordances are just as important as technical accuracy.

Prototype 3 focused on physical realism and immersion. The new drawer, shutter door, and lever interactions made the environment feel more tangible, and users appreciated the realistic motion. However, missing friction and guidance caused small usability issues, such as overshooting or uncertainty about interaction points. Participants asked for light resistance, haptic cues, and visual highlights to make movements more believable. This stage confirmed that realism must be balanced with clarity—strong physical feedback can enhance immersion only when it stays comfortable and predictable.

4. Evaluation of Aims

Testing Aim	Evaluation Outcome
Drawing and erasing interaction	Validated – Smooth, natural, and accurate.
Remote grab and floating tools	Strongly validated – Highest satisfaction, 4/5.
Physical reset button	Not achieved – Inconsistent sound and mechanical feedback.
Wrist menu	Partially achieved – Opened well, lacked functions.
Overall clarity and usability	Achieved – All tasks understood and completed.

Prototype 2 met most aims and confirmed that the XR workflow was functional and intuitive. Minor physical and feedback issues remained but did not block user performance.

5. Reflection

Prototype Session Review

Prototype 2 produced the most detailed user feedback.

Drawing and erasing worked smoothly, and the **remote grab** feature became the most successful element as it saved time and supported continuous creative flow.

However, the **physical reset button** had inconsistent sound and delayed response, reducing realism.

The **wrist menu** was easy to open but lacked active tools.

These issues led to refinements in the next stage.

In **Prototype 3**, new physical features such as drawers, levers, and gesture-based height control were added.

These elements provided stronger tactile and visual feedback, making interactions feel more realistic and comfortable.

Methodological Reflection

The step-by-step method—from desktop (IP1) to XR (IP2) to physical realism (IP3)—was effective for iterative design.

Each stage addressed a new layer: IP1 clarified workflow, IP2 tested immersive interaction, and IP3 enhanced feedback.

This structure supported focused improvement at every stage.

However, testing for sound and vibration was not always systematic; future sessions should include a checklist to ensure feedback consistency.

Concept Evaluation

The concept of a realistic XR creative tool was validated.

Users confirmed that natural gestures and tool freedom improved both control and creativity.

The **remote grab** proved central to efficiency, while the **reset button** showed that realism requires reliable response.

The **wrist menu** remained promising but unfinished, and Prototype 3 confirmed that embodied physical actions can greatly increase immersion when feedback is tuned correctly.

Improvements and Extensions

Next updates will focus on improving collision detection, consistent sound and vibration, smoother button mechanics, and a complete wrist menu with core tools.

Short tutorial prompts and highlight cues will guide first-time users.

Future development could explore **AR passthrough** for real-surface drawing and **multi-user collaboration** for shared creative spaces.

Appendix

- **GitHub Repository:** <https://github.com/YuchengChen-DECO7230/XR-Creative-Tool>
(Includes full documentation, testing data, and core functions for Prototype 1, 2 and 3.)

- **IP3 Test Questionnaires:**

Prototype 3 – Participant Questionnaire

Section 1. Task Evaluation

Please rate your experience with each interaction based on the following aspects:
1 = Very Poor 2 = Poor 3 = Neutral 4 = Good 5 = Excellent

Interaction Type	Realism (1-5)	Ease of Use (1-5)	Responsiveness (1-5)	Satisfaction (1-5)
Drawer (pull interaction)	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Shutter door (fling)	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Lever (scene switching)	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Camera height adjustment	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5

Section 2. Usability and Comfort

Please select or briefly describe your experience.

- Which cabinet interaction felt **more satisfying**?
☐ Drawer ☐ Shutter Door ☒ Both equally ☐ Neither
- Did the **lever feedback** (when switching scenes) feel clear and responsive?
☐ Yes ☒ Somewhat ☐ No
- Did you find the **camera height adjustment** intuitive and stable to use?
☐ Yes ☒ Somewhat ☐ No

Section 3. Immersion and Realism

- How **immersive** did the overall experience feel?
☒ Very immersive ☐ Somewhat immersive ☐ Neutral ☐ Slightly disconnected ☐ Not immersive
- Did the **cabinet and lever interactions** feel physically realistic?
☒ Yes ☐ Partly ☐ No
- Did the **scene switching** through the lever give a clear sense of environmental change?
☒ Yes ☐ Somewhat ☐ No
- Did adjusting the **camera height** enhance your feeling of control or presence in VR?
☐ Yes ☒ Somewhat ☐ No

Section 4. Open-ended Feedback

- Which feature did you **enjoy the most**, and why?
Camera height adjustment, make the experience personal & adaptable
- Which interaction felt **confusing or difficult**, and how could it be improved?
Drawer... a bit sensitive at times, add slight resistance

Thank you for your participation!

Your feedback will help us improve the realism, usability, and comfort of future XR creative environments.

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Shutter door (ding)	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5
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Section 4. Open-ended Feedback

- Which feature did you **enjoy the most**, and why?
Lever Smooth & Interesting
- Which interaction felt **confusing or difficult**, and how could it be improved?

Everything worked well, a fade change when switching scenes makes it better.
Immersive & intuitive. Responses and timing felt right.
More guidelines like highlight each interactive

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Section 4. Open-ended Feedback

- Which feature did you **enjoy the most**, and why?
Shutter Door, felt realistic, a good sense of physical motion
- Which interaction felt **confusing or difficult**, and how could it be improved?

Lever, scene changes happened suddenly need a smoother transition
Polished & Engaging Prototype
Add small Animations & Audio feedback during scene transitions -> cinematic & dynamic
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All Good, maybe shutter door a little bit smoother
Very Realistic... Easy to use, Well balanced
Add sound after scene change.

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Section 4. Open-ended Feedback

- Which feature did you **enjoy the most**, and why?
Drawer -> Natural and satisfying.
- Which interaction felt **confusing or difficult**, and how could it be improved?

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Change camera height more guidelines
Highly polished & immersive. Comfortable & easy to use.
Audio or haptic feedback

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Section 4. Open-ended Feedback

- Which feature did you **enjoy the most**, and why?
Drawer - Very natural, realistic sense of depth and movement!
- Which interaction felt **confusing or difficult**, and how could it be improved?